

Truth Values

- **Exercise 1**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$A : P \implies (Q \implies R)$$

$$B : (P \wedge Q) \implies R$$

are equivalent.

- **Solution 1.1**

We have

P	Q	R	$Q \implies R$	$P \wedge Q$	$P \implies (Q \implies R)$	$(P \wedge Q) \implies R$
1	1	1	1	1	1	1
1	1	0	0	1	0	0
1	0	1	1	0	1	1
1	0	0	1	0	1	1
0	1	1	1	0	1	1
0	1	0	0	0	1	1
0	0	1	1	0	1	1
0	0	0	1	0	1	1

Therefore, $A \equiv B$.

- **Example 2**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$A : (P \implies Q) \wedge (Q \implies R)$$

$$B : (\neg P \vee Q) \wedge (Q \vee R)$$

are equivalent.

- **Solution 2.2**

Case 1: A is false if and only if

$$(P \implies Q) \text{ is false or } (Q \implies R) \text{ is false}$$

which means

$$\left(\begin{array}{c} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{array} \right) \text{ or } \left(\begin{array}{c} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{array} \right)$$

Case 2: B is false if and only if

$$(\neg P \vee Q) \text{ is false or } (Q \vee R) \text{ is false}$$

which means

$$\left(\begin{array}{c} P \text{ is true} \\ \text{and} \\ Q \text{ is false} \end{array} \right) \text{ or } \left(\begin{array}{c} Q \text{ is false} \\ \text{and} \\ R \text{ is false} \end{array} \right)$$

In conclusion, both propositions are false exactly in the same cases.

Therefore, $A \equiv B$.

- **Example 3**

Let P, Q and R be three mathematical statements, we want to prove that the propositions

$$A : P \implies (Q \implies R)$$

$$B : (P \implies Q) \implies R$$

are not equivalent.

- **Solution 3.2**

if P is false, Q is true and R is false, then the proposition A is true and the proposition B is false, then A, B are not equivalent.

- **Example 4**

Are $A : \neg(P \wedge Q), B : \neg P \wedge \neg Q$ are equivalent?

- **Solution 4.1**

If P is true and Q is false, then A is true and B is false, then A, B are not equivalent.

- **Example 5**

Are $A : (P \implies Q) \wedge R, B : P \implies (Q \wedge R)$ are equivalent?

- **Solution 5.1**

If P is false, then B is always true. In this case, if R is false, A is false, then A, B are not equivalent.